

# Effectiveness of a Pharmacist-Led Intervention to Reduce Acid Suppression Therapy for Stress Ulcer Prophylaxis in ICUs in China: A Multicenter, Stepped-Wedge, Cluster-Randomized Controlled Trial

**OBJECTIVES:** This study aimed to evaluate the effectiveness of a pharmacist-led intervention in decreasing the overuse of stress ulcer prophylaxis (SUP) compared with the usual care for adult patients in Chinese ICUs.

**DESIGN:** Pragmatic, multicenter, stepped-wedge, cluster-randomized controlled trial.

**SETTING:** Twenty-six ICUs in China from October 2022 to March 2023.

**PATIENTS:** We enrolled 2199 patients 18 years old or older who were newly admitted to the participating ICUs.

**INTERVENTIONS:** Using the Medical Research Council framework for developing and evaluating complex intervention measures, a multidisciplinary team (Scenarios, Improving and Refining Interventions, Constructing, Refining and Testing Research Theories, Incorporating Stakeholders, Identifying Important Uncertainties, and Economics Considerations) designed a multifaceted intervention.

**MEASUREMENTS AND MAIN RESULTS:** The primary outcomes were the proportion of patients receiving SUP and that with overt gastrointestinal bleeding. We conducted intention-to-treat analyses using generalized linear mixed models to adjust for potential confounders (age, sex, and acute physiology and chronic health evaluation II score) with random effects for the site. The proportion of patients receiving SUP in the intervention group was lower than that in the control group (45.5% vs. 49.5%; odds ratio [OR], 0.81; 95% CI, 0.68–0.96;  $p = 0.017$ ). The proportion of patients with overt gastrointestinal bleeding was similar (3.7% vs. 4.0%; OR, 1.05; 95% CI, 0.65–2.85;  $p = 0.81$ ).

**CONCLUSIONS:** The pharmacist-led intervention reduced the proportion of patients receiving SUP in the ICUs, without significantly affecting the proportion of patients with overt gastrointestinal bleeding. These findings will help guide ICU medical decision-making.

**KEYWORDS:** cluster-randomized controlled trial; intensive care unit; multifaceted intervention; pharmacist; stress ulcer prophylaxis

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International guidelines have suggested acid suppressive therapy for stress ulcer prophylaxis (SUP) to prevent the potential progression of stress-related gastrointestinal mucosal damage to gastrointestinal bleeding (GIB) in patients admitted to ICUs (1). Previous cohorts and before and after studies demonstrated that approximately three of four patients received SUP during their ICU stay, with no identifiable risk of stress ulcer in 14.4–42% of them (2–6). A study in China reported a 12.9% overall rate for upper GIB within 14



## KEY POINTS

**Question:** What is the effectiveness of a pharmacist-led intervention in improving the rational use of stress ulcer prophylaxis (SUP) compared with the usual care for adult patients in Chinese ICUs.

**Findings:** This multicenter, stepped-wedge, cluster-randomized controlled trial found that the pharmacist-led intervention reduced the use of SUP in the ICU, without significantly affecting the proportion of patients with overt gastrointestinal bleeding.

**Meanings:** The evidence-based intervention by pharmacists and the ICU-SUP pathway designed in this study can guide clinical assessment for the proportion of appropriate SUP in the ICU.

days among adult patients in neurocritical care, with 79.0% receiving SUP with acid suppression (7).

Common SUP agents include proton pump inhibitors (PPIs) and histamine-2 receptor antagonists (H2RAs) for acid suppression therapy (AST) (5). PPIs are likely more effective for preventing clinically important GIB than H2RAs (odds ratio [OR], 0.38; 95% CI, 0.20–0.73) and placebo (OR, 0.24; 95% CI, 0.10–0.60) (all moderate quality evidence) (8); however, their use may lead to an increased risk of pneumonia (9). Some uncertainty regarding the impact of SUP on mortality and other in-hospital morbidities remains (9, 10). For patients with a low risk of GIB, the reduction in bleeding may be unimportant. Thus, SUP has very little impact on the absolute reduction in GIB. Further, the risk of acid suppression is likely to outweigh the benefits and leads to resource waste. Consequently, international organizations have called for the deprescription of PPIs in patients in the ICU (11, 12).

A recent systematic review showed that pharmacist-led interventions can reduce the proportion of patients receiving SUP inappropriately during ICU hospitalization and further reduce the occurrence of complications and save drug costs (13). However, the majority of relevant primary studies were retrospective, single-centered, and displayed poor confounding adjustment, with small sample sizes, limiting their ability to provide unbiased, precise, and generalizable findings.

Important changes have occurred within the pharmacy profession in the past few decades. During the 1950s and 1960s, a number of severe drug-related injuries prompted the recognition of the importance of medication safety. These events contributed to a shift in the role of the pharmacist from the traditional “drug-centered” approach to a “patient-centered” approach (14). Critical care pharmacists are essential members of the multiprofessional critical care team. A position statement outlined the recommended the roles of a critical care pharmacist and the scope of pharmacy services within the ICU (15). However, due to the different medical environment backgrounds, many of these recommendations were not applicable to Chinese pharmacists.

This study aimed to evaluate the effectiveness of a pharmacist-led intervention to improve the rational use of SUP compared with the usual care for adult patients in Chinese ICUs. The study hypothesized that evidence-based interventions by pharmacists would decrease the proportion of patients receiving SUP without increasing the proportion of patients with overt GIB in the ICU.

## MATERIALS AND METHODS

This study was a pragmatic, multicenter, stepped-wedge, cluster-randomized controlled trial. The report complies with the requirements of the Consolidated Standards of Reporting Trials (CONSORT) stepped-wedge cluster-randomized trials: extension of the CONSORT 2010 statement (CONSORT checklist in **Supplementary Material 2**, <http://links.lww.com/CCM/H671>) (16). We prospectively registered the trial in the Chinese Clinical Trials Registry (ChiCTR.org.cn, registration number: ChiCTR2200062588). This study was approved by the Ethics Committee of West China Second Hospital of Sichuan University (No. 2022175, date: July 15, 2022). The procedures followed adhered to the ethical standards of the Program for the Protection of Human Subjects and the Helsinki Declaration of 1975. All participating clinicians and patients provided written informed consent.

### Setting

We reached out to hospitals with ICUs and clinical pharmacists for participation via the Evidence-based

Pharmacy Committee of the Chinese Pharmaceutical Society. Inclusion criteria included: 1) the presence of a clinical pharmacist in the ICU, whether their responsibilities were confined to the ICU or extended to other clinical departments and 2) diverse ICU types, encompassing surgical ICUs (SICUs) (e.g., burn, neurosurgical, and thoracic ICUs), medical ICUs (e.g., respiratory, cardiovascular ICUs), and mixed ICUs (serving both surgical and medical patients). We did not apply any cluster-specific exclusion criteria.

## Study Population

We enrolled all patients 18 years old or older who were newly admitted to the enrolled ICUs. We excluded patients with appropriate prespecified indications for AST, including upper GIB, gastroesophageal reflux disease, peptic ulcer, mucosa-associated lymphoid tissue lymphoma, erosive esophagitis, gastroenteritis melanoma (Zollinger-Ellison syndrome), eosinophilic esophagitis, Barrett's esophagitis, and *Helicobacter pylori* infection, regardless of the status of patients receiving AST.

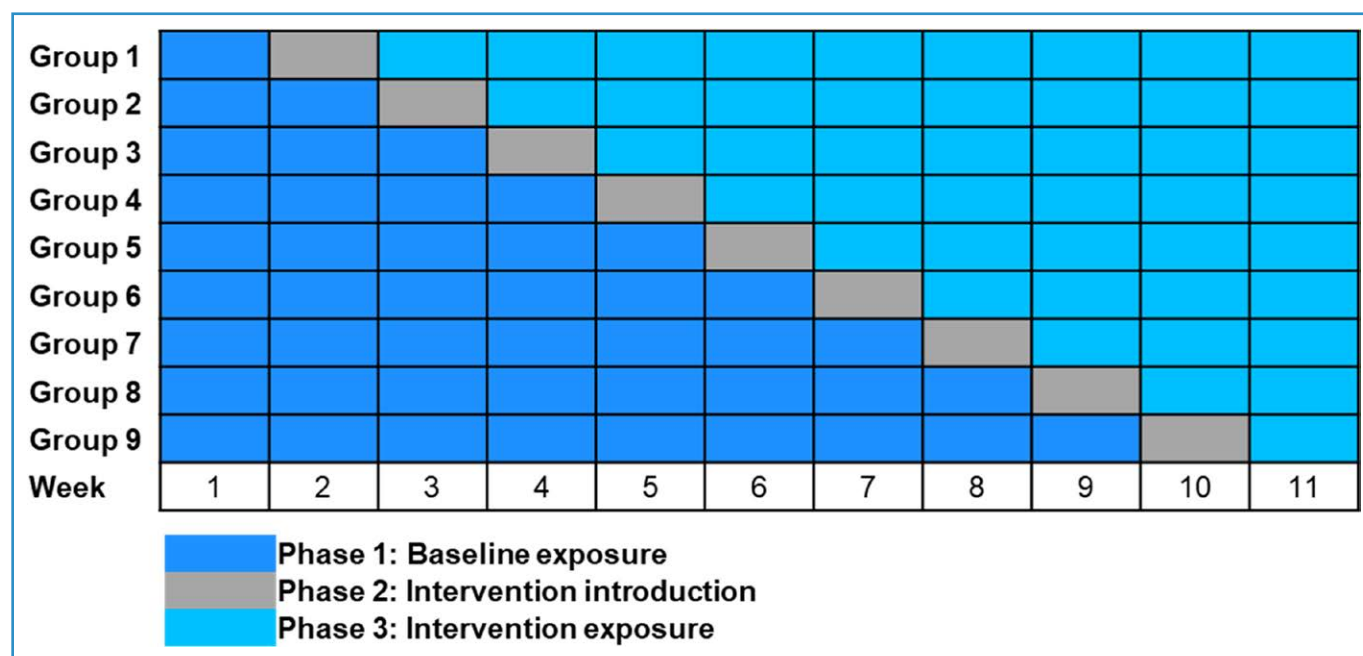
## Stepped Wedge Trial Design

To avoid contamination bias between ICUs or patients, or both, we defined each ICU and all patients within

that ICU as a cluster (17). We used a computer-generated random number list to allocate the 27 ICUs into nine sequences, each comprising three ICUs. Each ICU sequentially experienced three phases over 11 periods: 1) baseline phase (routine pharmacy services, control), with participants receiving usual care; 2) intervention introduction phase (the period to fully embed the pharmacist-led intervention; no patients were included in this period); and 3) intervention phase (all participants fulfilling the intervention condition were exposed to the pharmacist-led intervention). The duration of each period was at least 1 week. As participant enrollment was affected by the COVID-19 pandemic, certain ICUs extended the enrollment period to ensure sufficient sample sizes. After the tenth period, all ICUs had entered the “intervention phase” and stayed in this phase for the last week (Fig. 1). This trial was open-label, wherein clinicians, pharmacists, and participants were aware of the intervention.

## Interventions

The usual care in the 27 ICUs was consistent and included ward rounds (routinely rounding with the care team providing comprehensive recommendations), monthly retrospective prescription reviews, and medication recommendations to physicians on a specific clinical question. A monthly review encompassed retrospective assessment of the reasonableness

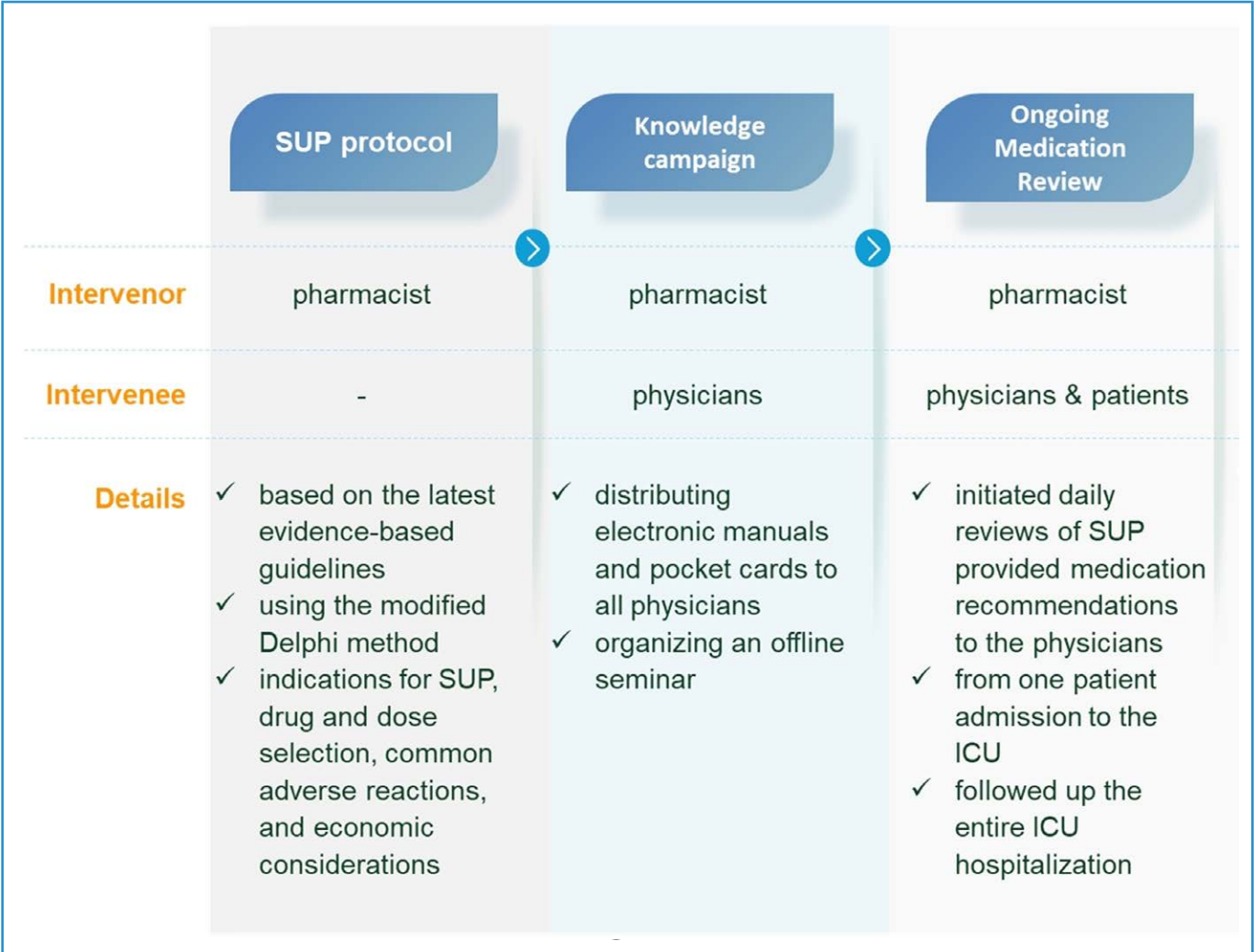


**Figure 1.** Stepped-wedge organization and timetable of controlled trial.

of medications used in the ICU over the past month, which may be for a class of drugs or all drugs for the same patient.

Using the Medical Research Council framework for developing and evaluating complex intervention measures (18), a multidisciplinary team (MDT) (Scenarios, Improving and Refining Interventions, Constructing, Refining and Testing Research Theories, Incorporating Stakeholders, Identifying Important Uncertainties, and Economics Considerations) designed a multifaceted intervention. This intervention comprised three components (Fig. 2): 1) Developing a manual for SUP prescription based on the latest evidence-based guidelines from the *British Medical Journal* (BMJ) rapid recommendation (19), using the modified Delphi method. This manual mainly considered the indications for SUP, drug and dose selection, common adverse reactions,

and economic considerations (**Supplementary Material 1**, <http://links.lww.com/CCM/H671>, for further details); 2) Distributing electronic manuals and pocket cards to all physicians on the medical team by pharmacists and organizing an offline seminar for drug knowledge dissemination and education activities; and 3) From admission to the ICU, pharmacists initiated daily reviews of SUP drug use, provided medication recommendations to the physicians based on the patient’s clinical condition and manual (**Supplementary Material 3**, <http://links.lww.com/CCM/H671>), and followed up the entire ICU hospitalization period. During the “intervention introduction phase,” all pharmacists underwent training from the study group and used uniform manuals, pocket cards, and slides. In the intervention phase, ICUs implemented a multifaceted intervention in addition to the usual care given by pharmacists.



**Figure 2.** The multifaceted intervention. SUP = stress ulcer prophylaxis.



## Outcomes

The trial assessed two co-primary outcomes: the proportion of patients receiving SUP (defined as the number of patients receiving SUP at any time during their ICU stay divided by the total intention-to-treat population) and those with overt GIB (defined as hematemesis; gross blood; or the presence of coffee-ground material in nasogastric tube aspirate, hematochezia, or melena) (20).

The secondary outcomes included the clinically important GIB, defined as overt bleeding complicated by any of the following within 24 hours after onset: 1) a spontaneous decrease in systolic blood pressure of greater than 20 mm Hg; 2) an orthostatic increase in heart rate of 20 beats/min and a decrease in systolic blood pressure of 10 mm Hg when the patient assumed an upright position; 3) a decrease in hemoglobin concentration of at least 2 g/dL and transfusion of 2 U of packed RBCs within 24 hours of bleeding; or 4) failure of the hemoglobin concentration to increase after transfusion by at least the number of transfused units minus 2 g/dL (20). Other secondary outcomes included adverse events related to SUP (*Clostridioides difficile* infection and hospital-acquired pneumonia, defined as pneumonia that occurs  $\geq 48$  hr after admission and not present at the time of admission) (21), length of hospital stay, all-cause mortality, the number of SUP-related recommendations provided by pharmacists for each patient, the duration of daily pharmacist reviews of patient conditions according to the SUP protocol and communication with physicians. Additionally, physician acceptance of pharmacist recommendations was categorized as full acceptance (complete medication regimen), partial acceptance (acceptance of discontinuation, medication, or change of medication, but not of a specific drug or dose), or rejection. Pharmacists evaluated events based on the protocol.

## Data Collection

At each study site, a pharmacist investigator collected hospital and ICU data using a standardized collection form in Microsoft Excel (Redmond, WA) at the beginning of the study. Throughout the study period, the local investigators prospectively collected daily patient data and documented pharmacists' practices using an electronic case report form within an Electronic Data Capture system until the end of the follow-up period.

## Sample Size and Power

This study hypothesized that evidence-based interventions by pharmacists would decrease the proportion of patients receiving SUP without increasing the proportion of patients with overt GIB in the ICU. First, we calculated the sample size according to the reduction in the proportion of patients receiving SUP (first co-primary outcome). Second, we calculated the power for detecting differences in the proportion of patients with overt GIB (second co-primary outcome) using the sample size defined for the first co-primary outcome (22), and a power greater than 0.8 was considered adequate (Supplementary Material 1, <http://links.lww.com/CCM/H671>, for more details).

The sample size for the two-group rate stepped-wedge randomized controlled trial was estimated using Power Analysis and Sample Size (PASS) 15.0.5 for SUP drug use (first co-primary outcome). A two-sided Wald Z test statistic was used, with the significance level of the test ( $\alpha$ ) set at 0.05, the intraclass correlation coefficient set at 0.010, and the test efficacy (power) set at 0.8; the design type was a complete step-wedge design, with a total of 27 clusters (K) and several transformed clusters per step (g) of 3. Based on prior research, the control rate ( $P_2$ ) was set at 0.72 (8), and the hypothesis that the pharmacist-led intervention could reduce SUP utilization by 10%, that is, the rate ratio ( $R_1$ ), was set to 0.9. The PASS calculations yielded a sample size of 2700 participants for ten study cycles (S), with nine transformations (T), several transformed clusters per step across three ICUs (g), and a sample size of ten cases per cluster per cycle (m).

Given the sample size of 2700, we calculated the power for testing the difference in the proportion of patients with overt GIB (second co-primary outcome) using the two groups of Poisson's rate stepped-wedge randomized controlled trials (22). In line with previous literature (23–25), the control rate for the occurrence of overt GIB ( $\lambda_2$ ) was set at 0.02, assuming a rate difference ( $D_1$ ) of 0.03. Other parameters remained constant, and the estimated power was 0.8281, which met the requirement for this study (power  $> 0.8$ ).

## Statistical Analyses

We performed the analyses within the intention-to-treat population. Baseline characteristics were described at both ICU and patient levels, assuming that the ICU baseline characteristics would remain consistent throughout the study period.

The proportion of patients receiving SUP and with overt GIB across different phases (groups) were described by frequencies and percentages. To control for systemic differences across these periods, we employed multivariable logistic regression through generalized linear mixed models (GLMMs), adjusted for calendar time and with random effects for each cluster (ICU) to analyze the primary outcomes. After adjusting for other potential confounders that were considered important prognostic factors, including age, sex, and Acute Physiology and Chronic Health Evaluation (APACHE II) score, the model included data from all periods except the intervention introduction phase.

GLMMs were used to compare the differences in the proportion of patients with GIB, significant SUP drug-related adverse events (hospital-acquired pneumonia, *C. difficile* infection), all-cause mortality, and length of ICU stay between groups.

All statistical tests were two-sided, with significance set at *p* value of less than 0.05. IBM SPSS software (Version v26.0; IBM Corp., Armonk, NY) was used for the statistical analyses.

## Missing Data

For instances where less than 5% of data were missing, we imputed the median values for the continuous variables and the mode for the categorical variables. When the missing data exceeded 5%, we used multiple imputations (23). Sensitivity analyses were performed based on complete case scenarios.

## RESULTS

Initially, 27 ICUs from 27 hospitals situated in 23 cities in China agreed to participate (see *Acknowledgments*). However, due to the COVID-19 epidemic, one ICU withdrew prior to data collection. The study enrolled 2592 patients from October 17, 2022, to March 31, 2023. Among them, 269 patients met the exclusion criteria and 124 refused to participate. The remaining 2199 patients were included in at least one analysis, with 1018 patients in the control phase and 1181 patients in the intervention phase (Fig. 3).

### ICU and Patient Characteristics

The included ICUs were all from tertiary class A (the most advanced and specialized institutions that

can provide comprehensive treatments and surgical operations) comprehensive teaching hospitals. Geographically, 11 ICUs were located in the east district (42.3%), ten in the central district (38.5%), and five in the west district (19.2%). These included 19 general ICUs (73.1%); two SICUs (7.7%); two cardiovascular ICUs (7.7%); and one neurosurgical ICU, one respiratory ICU, and one emergency ICU (3.8% each). The median number of ICU beds was 30 (interquartile range [IQR] = 16–42 beds), the median number of physicians was 17 (IQR = 11–24), and the median number of pharmacists was 1 (IQR = 1–2) (**Supplementary Material 4**, <http://links.lww.com/CCM/H671>).

The distribution of age, weight, sex, Glasgow Coma Scale scores, APACHE II scores, Sequential Organ Failure Assessment scores, history of gastrointestinal ulcer or bleeding within 1 year, history of nonsteroidal anti-inflammatory drugs use within 3 months, and present illness were similar between the two groups. More patients in the intervention group experienced severe trauma (5.5% vs. 4.9%) (**Table 1**).

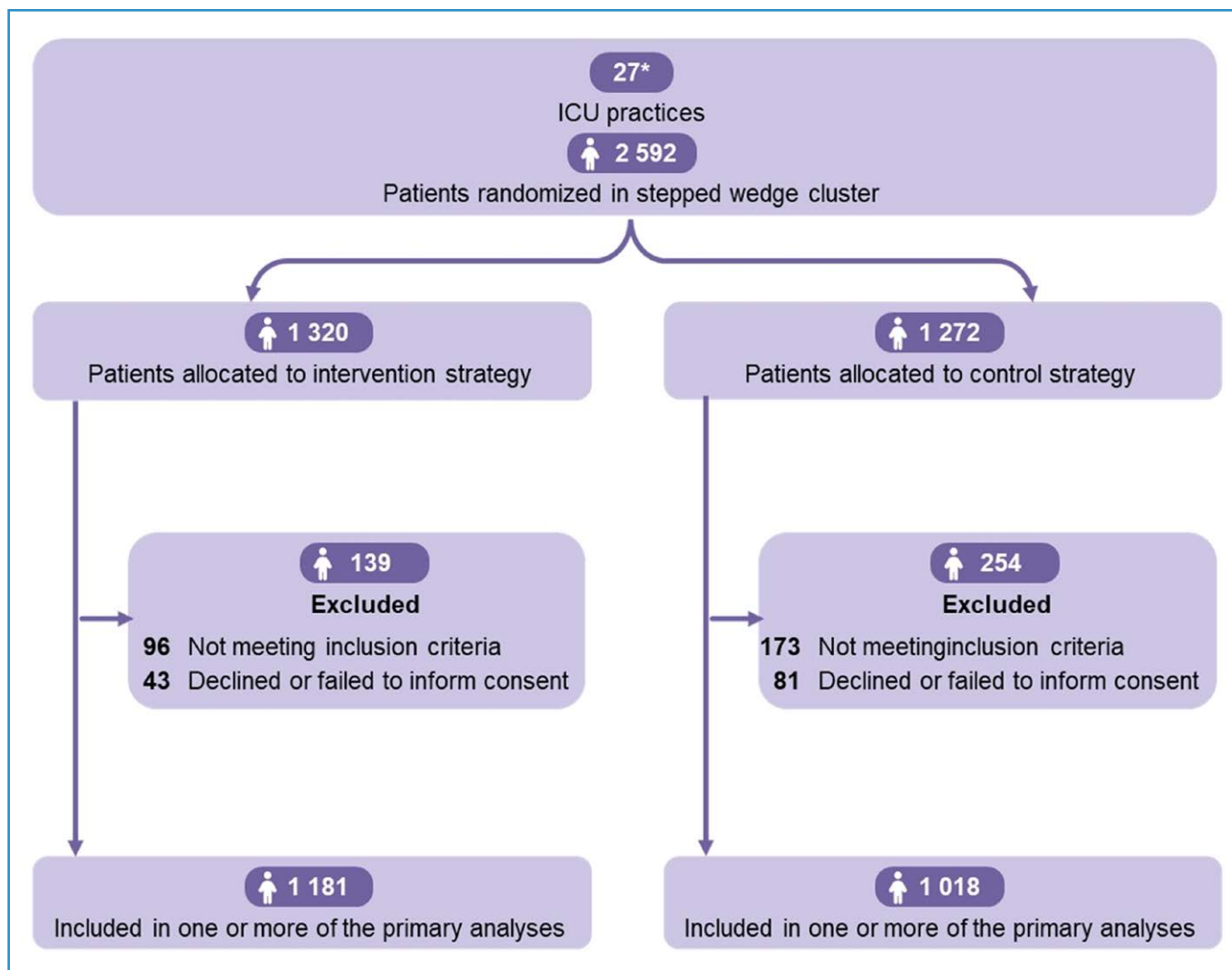
The ICU pharmacists had initiated their roles, which primarily encompass routine ward rounds, specific consultations, specialized prescription reviews within the ICU, and participation in MDT meetings. At baseline, a total of 13 hospitals (13/26, 50.0%) had established in-hospital ICU-SUP protocol. Among these, ten hospitals (10/13, 76.9%) had adopted the in-hospital ICU-SUP protocol as the criterion for the retrospective prescription drug review. Two hospitals (2/13, 15.4%) had integrated this in-hospital protocol into their information systems for physician reference, while one hospital (1/13, 7.7%) also enforced adherence to these standards by physicians (**Table 2**).

### Use of SUP

The proportion of patients receiving SUP was lower in the intervention group compared with that in the control group (45.5% vs. 49.5%; OR, 0.81; 95% CI, 0.68–0.96; *p* = 0.017) (**Table 3**).

### Clinical Outcomes of Enrolled Patients

The proportion of patients with overt GIB, clinically important GIB, hospital-acquired pneumonia, *C. difficile* infection, death, and length of ICU stay were similar during the intervention and control periods.



**Figure 3.** Flow chart of the controlled trial. \*Due to the COVID-19 pandemic, one ICU withdrew from the trial before inclusion started.

However, in each case, the CI around the relative effect was very wide (Table 3).

### Pharmacist Recommendations

Clinical pharmacists conducted 993 recommendations for 1181 patients. The pharmacists' recommendations were mainly focused on initiating SUP (593 times, 59.7%), and 60.1% of the pharmacists' recommendations (597 times) involving adjustments were fully or partially accepted by physicians (Table 4).

## DISCUSSION

In this study, we found that pharmacist-led interventions reduced the proportion of patients receiving SUP

in the ICU, while having no significant impact on the proportion of patients with overt GIB.

### Relation to Prior Work

Studies have evaluated the impact of pharmacists' intervention on rational drug use in SUP (2, 3, 13, 24–30). A recent systematic review revealed that a pharmacist-led intervention could reduce the proportion of patients receiving SUP inappropriately during ICU stays (13). Unlike the “irrational utilization rate” reported in previous studies, this trial assessed both the proportion of patients receiving SUP and the occurrence rate of adverse clinical outcomes. This approach was crucial due to the significant impact of the rationality criterion on research outcomes (13), and given the changes or updates in this criterion, the scientific

**TABLE 1.**  
**Baseline Characteristics of Participants**

| Characteristics                                                     | Intervention ( <i>n</i> = 1181) | Control ( <i>n</i> = 1018) |
|---------------------------------------------------------------------|---------------------------------|----------------------------|
| Age (yr)                                                            | 59.8 (17.7)                     | 60.4 (16.4)                |
| Weight (kg)                                                         | 65.9 (12.3)                     | 65.6 (12.5)                |
| Sex                                                                 |                                 |                            |
| Male                                                                | 754 (63.8%)                     | 657 (64.5%)                |
| Female                                                              | 427 (36.2%)                     | 361 (35.4%)                |
| Severity scores                                                     |                                 |                            |
| Glasgow Coma Scale                                                  | 11.0 (4.0–15.0)                 | 11.0 (7.0–15.0)            |
| Acute Physiology and Chronic Health Evaluation II                   | 14.0 (10.0–19.0)                | 14.0 (9.0–19.0)            |
| Sequential Organ Failure Assessment score                           | 4.0 (3.0–6.0)                   | 4.0 (2.0–6.0)              |
| History of gastrointestinal ulcer/bleeding within the past 1 yr     |                                 |                            |
| Yes                                                                 | 15 (1.3%)                       | 16 (1.6%)                  |
| Missing                                                             | 1 (0.1%)                        | 7 (0.7%)                   |
| Nonsteroidal anti-inflammatory drugs medication history within 3 mo |                                 |                            |
| Yes                                                                 | 59 (5.0%)                       | 57 (5.6%)                  |
| Missing                                                             | 1 (0.1%)                        | 16 (1.6%)                  |
| Present illness                                                     |                                 |                            |
| Mechanical ventilation without enteral nutrition                    | 462 (39.1%)                     | 361 (35.5%)                |
| Shock                                                               | 160 (13.5%)                     | 183 (18.0%)                |
| Severe, multiple trauma                                             | 65 (5.5%)                       | 50 (4.9%)                  |
| Severe craniocerebral injury                                        | 56 (4.7%)                       | 68 (6.7%)                  |
| Coagulopathy                                                        | 38 (3.2%)                       | 35 (3.4%)                  |
| Spinal cord injury                                                  | 21 (1.8%)                       | 16 (1.6%)                  |
| Tumor chemotherapy                                                  | 19 (1.6%)                       | 39 (3.8%)                  |
| Severe burn                                                         | 0 (0.0%)                        | 1 (0.1%)                   |

Data are expressed as *n* (%), mean (sd), and median (interquartile range).

interpretation and application of the research results are rather challenging. Therefore, this study implemented rational drug use principles for clinical drug application and the clinical benefit of patients. Because the *BMJ* rapid recommendation guidelines provided weak (conditional) recommendations for initiating SUP in high-risk groups (19), this study suggests that reducing the proportion of patients receiving SUP may be achievable without statistically significant increases in the proportion of patients with clinical adverse outcomes. This trial found that the proportion of patients receiving SUP after pharmacists' intervention was lower than that of the control group (45.5% vs. 49.5%), albeit modest. However, there was no statistically significant difference in the proportion of patients with

overt GIB after the pharmacists' interventions. This was similar to the findings of Hammond et al (2) in a single-center, retrospective cohort study, which demonstrated that pharmacists' evidence-based interventions reduced the irrational use of SUP drugs without increasing the occurrence rate of GIB.

### Implications for Future Study and Practice

The pharmacist-led intervention designed in this study can be extended to a wider range of healthcare institutions, incorporating the SUP risk stratification and the pathway developed in this study to guide ICU medical decision-making. Combined with electronic information, it serves as an auxiliary means to optimize



**TABLE 2.**  
**Baseline Characteristic of ICUs' Pharmacists**

| ICUs  | ICU Pharmacist Usual Care |              |                             |                                                | SUP-Related |                         |
|-------|---------------------------|--------------|-----------------------------|------------------------------------------------|-------------|-------------------------|
|       | Ward Round                | Consultation | Multidisciplinary Treatment | Periodic Prescription Review (Retrospectively) | Protocol    | Implementation          |
| SUP02 | Yes                       | Yes          | –                           | –                                              | –           | –                       |
| SUP03 | Yes                       | Yes          | –                           | –                                              | –           | –                       |
| SUP06 | Yes                       | Yes          | Yes                         | –                                              | –           | –                       |
| SUP08 | Yes                       | Yes          | –                           | –                                              | Yes         | Review criteria         |
| SUP09 | Yes                       | Yes          | Yes                         | –                                              | –           | –                       |
| SUP10 | Yes                       | Yes          | Yes                         | –                                              | –           | –                       |
| SUP12 | Yes                       | Yes          | –                           | –                                              | –           | –                       |
| SUP15 | Yes                       | Yes          | Yes                         | –                                              | –           | –                       |
| SUP17 | Yes                       | –            | Yes                         | Yes                                            | –           | –                       |
| SUP18 | Yes                       | Yes          | Yes                         | –                                              | Yes         | System tools, tips      |
| SUP19 | Yes                       | –            | –                           | –                                              | Yes         | Review criteria         |
| SUP20 | –                         | Yes          | –                           | –                                              | Yes         | System tools, mandatory |
| SUP21 | Yes                       | Yes          | Yes                         | –                                              | Yes         | Review criteria         |
| SUP22 | Yes                       | Yes          | Yes                         | –                                              | Yes         | Review criteria         |
| SUP23 | Yes                       | Yes          | –                           | –                                              | Yes         | Review criteria         |
| SUP25 | Yes                       | Yes          | Yes                         | –                                              | Yes         | Review criteria         |
| SUP27 | Yes                       | –            | –                           | –                                              | –           | –                       |
| SUP28 | Yes                       | Yes          | –                           | –                                              | Yes         | Review criteria         |
| SUP29 | Yes                       | Yes          | Yes                         | –                                              | Yes         | Review criteria         |
| SUP30 | Yes                       | Yes          | –                           | –                                              | –           | –                       |
| SUP31 | Yes                       | Yes          | –                           | –                                              | Yes         | Review criteria         |
| SUP32 | Yes                       | Yes          | –                           | –                                              | –           | –                       |
| SUP33 | Yes                       | Yes          | –                           | –                                              | –           | –                       |
| SUP34 | Yes                       | Yes          | –                           | –                                              | Yes         | Review criteria         |
| SUP35 | Yes                       | Yes          | Yes                         | Yes                                            | Yes         | System tools, tips      |
| SUP37 | Yes                       | –            | –                           | –                                              | –           | –                       |

SUP = stress ulcer prophylaxis.

the implementation details of the intervention, which further improves its applicability and operability, thus promoting rational drug use in clinical settings. Additionally, a preliminary study was conducted to explore the risk factors for GIB to support the development of Chinese guidelines for SUP in the ICU.

### Strengths

Compared with the predominantly retrospective observational existing pharmacist-led studies, this

study adopted a stepped-wedge, multicenter, randomized controlled trial design to verify the impact of pharmacist-led evidence-based intervention on compound outcomes (the proportion of patients receiving SUP and patient clinical outcomes), and the results yielded higher representativeness and generalizability. Each unit served as its control and intervention, introduced at separate times according to randomization. Furthermore, the analysis was adjusted for the most important patient characteristics, further enhancing the study's robustness. Thus,

**TABLE 3.**  
**Stress Ulcer Prophylaxis Drug Utilization Rate and Clinical Outcomes of Enrolled Patients**

| Outcome Index                                           | Intervention<br>(n = 1181) | Control<br>(n = 1018) | Effect Estimates         |       |
|---------------------------------------------------------|----------------------------|-----------------------|--------------------------|-------|
|                                                         | n (%)                      | n (%)                 | OR (95% CI)              | p     |
| Stress ulcer prophylaxis drugs use                      | 537 (45.5)                 | 504 (49.5)            | 0.81 (0.68–0.96)         | 0.017 |
| Overt gastrointestinal bleeding                         | 44 (3.7)                   | 41 (4.0)              | 1.05 (0.65–2.85)         | 0.81  |
| Clinically important bleeding                           | 13 (1.1)                   | 8 (0.8)               | 1.09 (0.68–1.74)         | 0.72  |
| Hospital-acquired pneumonia                             | 78 (6.6)                   | 74 (7.3)              | 1.19 (0.86–1.64)         | 0.29  |
| <i>Clostridioides difficile</i> infection               | 2 (0.2)                    | 7 (0.7)               | 0.94 (0.58–1.53)         | 0.80  |
| All-cause mortality                                     | 68 (5.8)                   | 67 (6.6)              | 1.00 (0.53–1.40)         | 0.99  |
| Length of ICU stay, hr, median<br>(interquartile range) | 100.0 (49.5–186.0)         | 100.0 (47.0–198.3)    | –0.061 (–0.547 to 0.424) | 0.80  |

OR = odds ratio.

**TABLE 4.**  
**Recommendations of Pharmacist**

| Recommendation     | n (%)      |
|--------------------|------------|
| Initiate SUP       | 593 (59.7) |
| Partial acceptance | 301 (30.3) |
| Reject             | 230 (23.2) |
| Full acceptance    | 62 (6.2)   |
| Stop SUP           | 209 (21.0) |
| Reject             | 93 (9.4)   |
| Full acceptance    | 83 (8.4)   |
| Partial acceptance | 33 (3.3)   |
| Adjust SUP         | 191 (19.2) |
| Reject             | 73 (7.4)   |
| Partial acceptance | 65 (6.5)   |
| Full acceptance    | 53 (5.3)   |

SUP = stress ulcer prophylaxis.  
Full acceptance: complete medication regimen; partial acceptance: acceptance of discontinuation, medication, or change of medication, but not of a specific drug or dose; reject.

this study bears a much lower risk of bias than previous observational studies.

**Limitations**

Pharmacists’ interventions mainly involved pharmacist-led education and recommendations, with a relatively weak implementation intensity compared with mandatory or administrative interventions. The diagnostic

criterion for stress ulcers was visible ulcers identified through gastroscopy. Considering the feasibility and clinical significance of employing invasive procedures in critically ill patients and not the standard of care, this study focused on the progression of GIB as a primary outcome rather than the occurrence rate of stress ulcers, leading to some level of indirection. Due to the imbalance of pharmacist-bed ratio in Chinese medical institutions, it was difficult for pharmacists in some centers to cover all ICU beds, so they selected a fixed number of beds or medical groups for patient enrollment, which may cause some bias. Additionally, considering the feasibility of intervention implementation, this study only included third-level A teaching hospitals with clinical pharmacists possessing ICU work experience. Whether the trial conclusions and intervention measures can be applied to other hospital types and pharmacists requires further exploration.

The timing of our research coincided with the COVID-19 pandemic, and the strict measures imposed by the Chinese government unfortunately resulted in our study falling short of the intended sample size. This limitation likely reduced the statistical power of our study, increasing the risk of failing to detect a true effect and affecting the generalizability of our conclusions. The resulting variability could have obscured precise effect estimation, impacting the reproducibility of our findings. Furthermore, the reduced sample size constrained our capacity for subgroup analysis, thus limiting insights into demographic or clinical effect modifiers. These constraints caution

against over-reliance on our study for clinical guidance and may have influenced future research funding and priorities. Nonetheless, we presented our findings as a valuable contribution to the field, advocating for larger-scale studies to validate and expand upon our work.

## CONCLUSIONS

The pharmacist-led intervention reduced the proportion of patients receiving SUP in the ICUs. Meanwhile, its impact on the proportion of patients with overt GIB was not significant. The evidence-based intervention by pharmacists and the ICU-SUP pathway designed in this study can guide clinical assessment for appropriate SUP in the ICU.

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Drs. Li and Zeng were involved in conceptualization, development and design of methodology, analysis and synthesis of study data, preparation, creation, and presentation of the published work. Dr. Xu was involved in conceptualization, preparation, creation, and presentation of the published work. Drs. Olsen and Granholm were involved in conceptualization, critical review, commentary, and revision of the published work. Drs. Zhang and Guyatt were involved in conceptualization, critical review, commentary, and revision of the published work, management and coordination of the research activity, and planning and execution.

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The study protocol was approved by the Ethics Committee of the West China Second University and all participating hospitals. Informed consent was provided by all participating physicians and patients (No. 2022175).

All data generated or analyzed during this study could be accessed by contacting the corresponding author.

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## REFERENCES

- Evans L, Rhodes A, Alhazzani W, et al: Surviving sepsis campaign: International guidelines for management of sepsis and septic shock 2021. *Crit Care Med* 2021; 49:e1063–e1143
- Hammond DA, Killingsworth CA, Painter JT, et al: Impact of targeted educational interventions on appropriateness of stress ulcer prophylaxis in critically ill adults. *Pharm Pract* 2017; 15:948
- Buckley MS, Park AS, Anderson CS, et al: Impact of a clinical pharmacist stress ulcer prophylaxis management program on inappropriate use in hospitalized patients. *Am J Med* 2015; 128:905–913
- Masood U, Sharma A, Bhatti Z, et al: A successful pharmacist-based quality initiative to reduce inappropriate stress ulcer prophylaxis use in an academic medical intensive care unit. *Inquiry* 2018; 55:46958018759116
- Frاندah W, Colmer-Hamood J, Nugent K, et al: Patterns of use of prophylaxis for stress-related mucosal disease in patients admitted to the intensive care unit. *J Intensive Care Med* 2014; 29:96–103
- Farrell CP, Mercogliano G, Kuntz CL: Overuse of stress ulcer prophylaxis in the critical care setting and beyond. *J Crit Care* 2010; 25:214–220
- Wei J, Jiang R, Li L, et al: Stress-related upper gastrointestinal bleeding in adult neurocritical care patients: A Chinese multicenter, retrospective study. *Curr Med Res Opin* 2019; 35:181–187
- Alhazzani W, Alshamsi F, Belley-Cote E, et al: Efficacy and safety of stress ulcer prophylaxis in critically ill patients: A network meta-analysis of randomized trials. *Intensive Care Med* 2018; 44:1–11
- Wang Y, Ye Z, Ge L, et al: Efficacy and safety of gastrointestinal bleeding prophylaxis in critically ill patients: Systematic review and network meta-analysis. *BMJ* 2020; 368:l6744
- Reynolds PM, Wells L, Powell M, et al: Associated mortality risk of proton pump inhibitor therapy for the prevention of stress ulceration in intensive care unit patients: A systematic review and meta-analysis. *J Clin Gastroenterol* 2022; 57:586–594
- Targownik LE, Fisher DA, Saini SD: AGA clinical practice update on de-prescribing of proton pump inhibitors: Expert review. *Gastroenterology* 2022; 162:1334–1342
- Farrell B, Lass E, Moayyedi P, et al: Reduce unnecessary use of proton pump inhibitors. *BMJ* 2022; 379:e069211
- Xu P, Yi Q, Wang C, et al: Pharmacist-led intervention on the inappropriate use of stress ulcer prophylaxis pharmacotherapy in intensive care units: A systematic review. *Front Pharmacol* 2021; 12:741724
- Anderson S: The state of the world's pharmacy: A portrait of the pharmacy profession. *J Interprof Care* 2002; 16:391–404
- Lat I, Paciullo C, Daley MJ, et al: Position paper on critical care pharmacy services: 2020 update. *Crit Care Med* 2020; 48:e813–e834
- Hemming K, Taljaard M, McKenzie JE, et al: Reporting of the CONSORT extension for stepped-wedge cluster randomised trials: Extension of the CONSORT 2010 statement with explanation and elaboration. *BMJ* 2018; 363:k1614
- Torgerson DJ: Contamination in trials: Is cluster randomisation the answer? *BMJ* 2001; 322:355–357
- Skivington K, Matthews L, Simpson SA, et al: A new framework for developing and evaluating complex interventions: Update of Medical Research Council guidance. *BMJ* 2021; 374:n2061
- Ye Z, Reintam Blaser A, Lytvyn L, et al: Gastrointestinal bleeding prophylaxis for critically ill patients: A clinical practice guideline. *BMJ* 2020; 368:l6722
- Cook DJ, Fuller HD, Guyatt GH, et al: Risk factors for gastrointestinal bleeding in critically ill patients. Canadian Critical Care Trials Group. *N Engl J Med* 1994; 330:377–381
- American Thoracic Society; Infectious Diseases Society of America: Guidelines for the management of adults with hospital-acquired, ventilator-associated, and healthcare-associated pneumonia. *Am J Respir Crit Care Med* 2005; 171:388–416
- Hemming K, Taljaard M: Sample size calculations for stepped wedge and cluster randomised trials: A unified approach. *J Clin Epidemiol* 2016; 69:137–146
- van Buuren S: *Flexible Imputation of Missing Data*. Boca Raton, FL, CRC Press, 2018
- Coursol CJ, Sanzari SE: Impact of stress ulcer prophylaxis algorithm study. *Ann Pharmacother* 2005; 39:810–816
- Wohlt PD, Hansen LA, Fish JT: Inappropriate continuation of stress ulcer prophylactic therapy after discharge. *Ann Pharmacother* 2007; 41:1611–1616
- Zeigler AJ, McAllen KJ, Slot MG, et al: Medication reconciliation effect on prolonged inpatient stress ulcer prophylaxis. *Ann Pharmacother* 2008; 42:940–946
- Hatch JB, Schulz L, Fish JT: Stress ulcer prophylaxis: Reducing non-indicated prescribing after hospital discharge. *Ann Pharmacother* 2010; 44:1565–1571
- Tasaka CL, Burg C, Vanosdol SJ, et al: An interprofessional approach to reducing the overutilization of stress ulcer prophylaxis in adult medical and surgical intensive care units. *Ann Pharmacother* 2014; 48:462–469
- Masood U, Sharma A, Bhatti Z, et al: A successful pharmacist-based quality initiative to reduce inappropriate stress ulcer prophylaxis use in an academic medical intensive care unit. *Inquiry* 2018; 55:S445
- Anstey MH, Litton E, Palmer RN, et al: Clinical and economic benefits of de-escalating stress ulcer prophylaxis therapy in the intensive care unit: A quality improvement study. *Anaesth Intensive Care* 2019; 47:503–509